Design and Development of a Relational SQLite Database

UK Electric Vehicle Charging Behaviour (2025–2026) | MSc Data Science – SQL Assignment

Student name Nithin Kumar Galla	**Student ID** 25011392
GitHub repository Paste repository URL here: https://github.com/nithin522/data_mining_assignment	**Module / tutor** SQL Database Assignment | David Lagattuta

1. Purpose and scope

This project designs a relational SQLite database representing activity across a simulated UK public electric-vehicle (EV) charging network. EV charging is a useful data-science context because it combines transport infrastructure, energy consumption, pricing and user behaviour. The database supports queries about charging demand, charger characteristics, session outcomes and data quality. **All records are synthetically generated**; they do not describe real drivers, charging sessions or commercial operators.

Project objectives

- Design a normalised relational schema separating regions, stations, chargers, vehicles, tariffs and chargingtransactions.
- Demonstrate primary keys, foreign keys, composite keys and attribute-level constraints.
- Generate more than 1,000 realistic-looking records using reproducible Python code.
- Include controlled data-quality issues and show how they can be identified during validation.

2. Data generation and realism

The database is generated in Python using the standard **random**, **datetime** and **sqlite3** libraries. A fixed random seed (20261105) makes generation reproducible. The generator creates 8 UK regions, 40 stations, 160 chargers, 350 pseudonymous vehicles, 48 tariff combinations, 2,000 charging sessions and 10,000 session events. Weighted choices and bounded random distributions model connector mix, charger status, session outcomes, energy, duration, battery state and temperature. Generated coordinates are illustrative and must not be used for navigation.

Generation choices and scope

The simulation uses plausible categories and bounded values rather than claiming to reproduce the full UK charging market. Session dates span 2025–2026; energy, duration and battery percentages are generated within defined ranges. This supports SQL joins, grouping, aggregation and data-quality checks, but it is not a statistically representative sample of actual charging activity. No third-party dataset or real personal information is used.

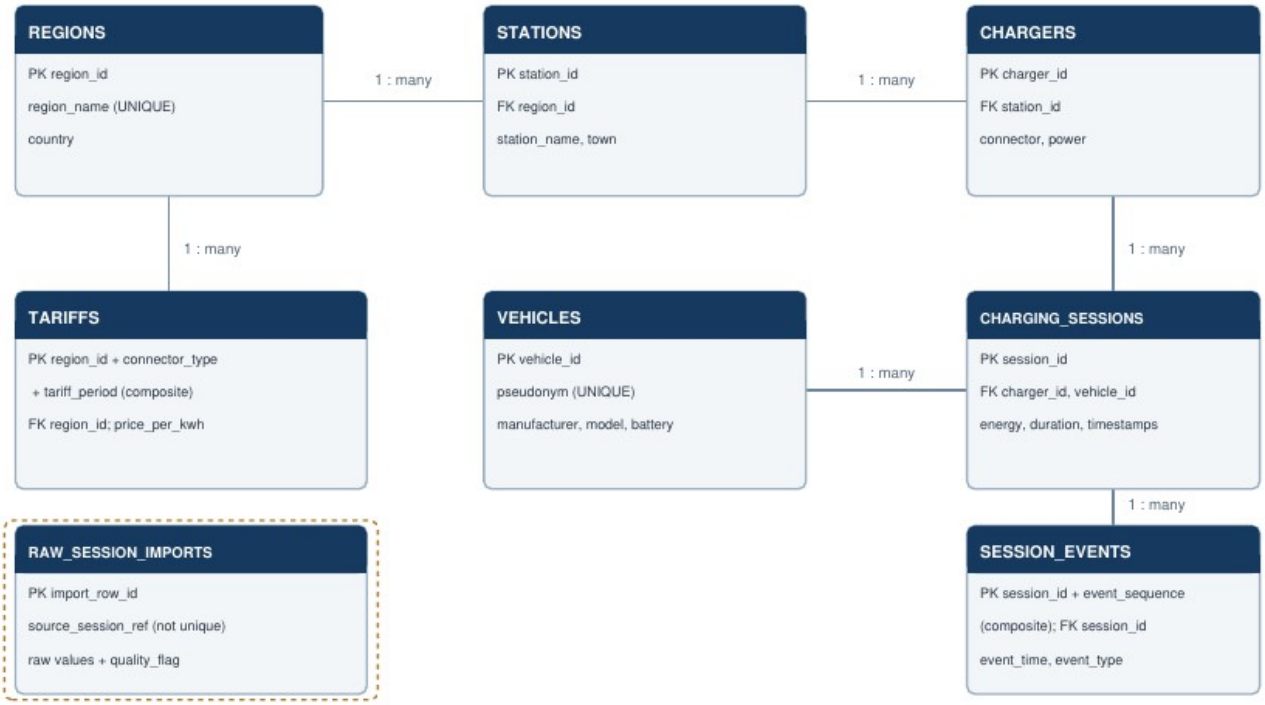
3. Database contents

Table	Rows	Purpose
regions	8	UK regional reference data
stations	40	Charging locations within regions
chargers	160	Individual charging units
vehicles	350	Synthetic EV characteristics
tariffs	48	Regional connector prices by period
charging_sessions	2,000	Main charging transaction table
session_events	10,000	Ordered events for each session
raw_session_imports	2,012	Unvalidated source-feed staging records

Design rationale. Reference entities are separated from transaction records to reduce repeated values and update anomalies. For example, a station's region is stored once rather than repeated in every charging session.

4. Entity relationship diagram (ERD)

The diagram shows the principal one-to-many relationships. PK means primary key and FK means foreign key. The tariffs and session_events tables contain composite primary keys. The raw import staging table is deliberately separate because it contains unvalidated source data.



Relationship interpretation

Parent	Child	Relationship
regions	stations	One region can contain many stations.
stations	chargers	One station can contain many chargers.
regions	tariffs	One region has several connector and tariff-period combinations.
chargers	charging_sessions	One charger can be referenced by many sessions.
vehicles	charging_sessions	One vehicle can participate in many sessions.
charging_sessions	session_events	One session can contain multiple ordered events.

The **tariffs** table uses (region_id, connector_type, tariff_period) as a composite primary key because the combination identifies one tariff. The **session_events** table uses (session_id, event_sequence) so an event sequence number is unique within each session. The **raw_session_imports** table has an import_row_id primary key but intentionally does not enforce uniqueness on source_session_ref, allowing repeated feed messages to be investigated.

Why the staging table is separate

Raw imported data is kept apart from curated operational tables so that incomplete or repeated feed messages can be inspected without weakening the integrity of the main schema. After validation, accepted records can be transformed and inserted into the charging_sessions table. This separation reflects a common data-engineering pattern: retain source evidence, validate it, then publish clean records for analysis.

5. Measurement types and attribute constraints

Scale	Example attribute	Justification
Nominal	connector_type (CCS, CHAdeMO, Type 2)	Categories without inherent order.
Ordinal	accessibility_rating (Poor–Excellent)	Categories have a meaningful order.
Interval	ambient_temp_c	Celsius has an arbitrary zero point.
Ratio	energy_kwh, duration_minutes, battery_capacity_kwh	Meaningful zero; ratios can be compared.

Integrity constraints

Primary keys uniquely identify records. Foreign keys preserve valid links between regions, stations, chargers, vehicles, sessions and events. NOT NULL constraints protect essential fields; UNIQUE constraints prevent repeated region names, pseudonyms and charger labels within a station. CHECK constraints restrict categories and plausible numeric ranges: energy must be positive, charger power must be within its defined range, and battery percentages must remain between 0 and 100. Optional model years and event notes may be NULL because unknown or unrecorded values are plausible in operational data.

6. Data quality and validation

The raw_session_imports staging table deliberately contains missing energy values, missing payment references and 12 duplicate source-feed records. Rows are labelled Clean, Missing field or Duplicate. The staging table does not impose uniqueness on source_session_ref because repeated messages can occur when an upstream service retries a transmission. The curated charging_sessions table is separate and protected by a primary key and foreign keys. Validation confirmed 2,000 charging sessions and no foreign-key violations. The notebook checks table counts, missing values and repeated source references.

Validation approach

Validation is performed at more than one level. Structural checks confirm that expected tables exist and that primary-key and foreign-key rules are respected. Content checks inspect NULL values, duplicate source references and values outside expected ranges. The distinction between the staging and curated tables is important: a duplicate in a raw feed is a data-quality finding, not necessarily a duplicate transaction in the final analytical table.

7. Ethical and data privacy considerations

Charging histories can reveal routines, commuting patterns or visits to sensitive locations. This project therefore uses synthetic records and generated pseudonyms, not real names, registration plates, phone numbers, payment credentials or home addresses. In a real deployment, access to session-level records should be role-restricted, retention periods should be defined, and only necessary data should be collected. Location and timestamp information should be aggregated or generalised before wider reporting because combinations of these fields can still identify individuals. Payment processing should be handled by an appropriate provider; this database stores only a broad payment-method category and a synthetic staging reference. Any real-world implementation would require an appropriate lawful basis, transparent privacy information and suitable security controls under applicable UK data-protection requirements.

8. Conclusion

The database demonstrates a multi-table relational design, all four measurement scales, foreign-key links, two composite primary keys, realistic constraints and deliberate data-quality defects. The Python generator and annotated notebook make the work reproducible and include example SQL analysis using joins, aggregation and time-based grouping. Because the records are synthetic, the results demonstrate database techniques rather than actual UK EV charging behaviour.

Accompanying files: *generate_ev_database.py* (generator), *uk_ev_charging_database.ipynb* (annotated notebook), *uk_ev_charging_2026.sqlite* (SQLite database).